

Integration by Parts

Pick a topic and answer each step before it's revealed.

Level

Easy

Medium

Hard

Type (suggested order 1 → 6)

1. Algebraic $\times$ exp/trig

2. Logarithms

3. Inverse trig

4. Repeated parts

5. Cyclic ($e^x \sin x$)

6. Definite

Evaluate $\int x \sin x \, dx$

Choose u and dv (LIATE: Log, Inverse trig, Algebraic, Trig, Exp):

$u = 1, dv = x \sin x \, dx$

$u = \sin x, dv = x \, dx$

$u = x, dv = \sin x \, dx$

$u = x \sin x, dv = dx$

 [Explain this step](#)

